The invariant field is the base produced by the action

SOURCE / Artin fixed-field theorem

$$F = \mathbb{Q}(x_g : g \in G)$$

fixed field

$$K = F^G, \qquad \mathrm{Gal}(F/K) \cong G$$

Correct realization over the invariant base

The base is not yet $\mathbb{Q}$

$\mathrm{trdeg}_{\mathbb{Q}} K = |G|$ because F/K is finite.